MCA SOFTWARE ENGINEERING LAB

Experiment No. 3: Use Case Diagram for Shield AI

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Aim

To identify the actors and use cases of Shield AI and prepare a use case diagram for a system that analyzes text from emails, SMS messages, and other sources to identify potential scams.

Step 1: Actors

Actor	Type	Reason
User	Primary	Pastes or enters suspicious text and reviews the analysis result.
Email / SMS Source	External input source	Provides message text for analysis; the user may copy text from these sources.
Shield AI Analysis Engine	Internal system component	Analyzes text and produces a risk classification and explanation.

The email or SMS itself is a data source, not a human actor. The user initiates scanning and receives the result from Shield AI.

Step 2: Use Cases

No.	Use case	Actor(s)
UC-01	Enter or Paste Message Text	User
UC-02	Select Message Source Type	User
UC-03	Validate Input Text	User
UC-04	Preprocess and Clean Text	Shield AI
UC-05	Analyze Scam Indicators	Shield AI
UC-06	Classify Scam Risk Level	Shield AI
UC-07	View Detection Result and Warning	User

No.	Use case	Actor(s)
UC-08	View Explanation / Detected Indicators	User
UC-09	View Scan History (if enabled)	User
UC-10	Submit Feedback on Result	User
UC-11	Clear Text and Start New Scan	User
UC-12	Handle Empty or Invalid Input	Shield AI
UC-13	Review Safety Guidance	User

Step 3: Use Case Diagram

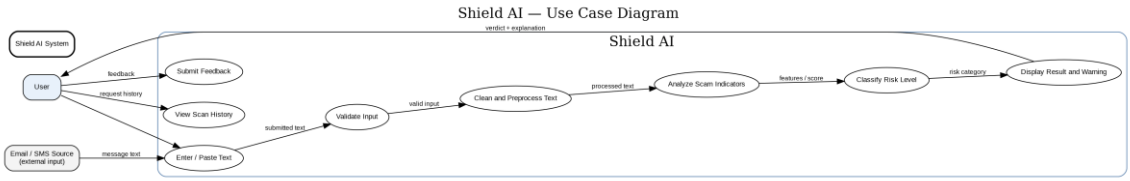


Figure 1: Use case diagram of Shield AI

Include relationships

The user submits text for scanning. Shield AI validates and preprocesses the text, analyzes scam indicators, classifies the risk, and displays a warning with an explanation. The user may review prior scans if history is implemented and provide feedback.

Check

- Actors are outside the system boundary and use cases are inside.
- Every use case is connected to an actor.
- The use cases are based on the proposed Shield AI workflow. History and feedback functions are optional and should be included only if implemented in the final application.

Result

In this experiment, we identified the user and the main use cases of Shield AI and prepared a use case diagram showing how message text is submitted, analyzed, classified, and presented as a potential scam warning.